

Non-invasive genetic sampling of sympatric marten species (*Martes martes* and *Martes foina*): assessing species and individual identification success rates on faecal DNA genotyping

Aritz Ruiz-González · María José Madeira · Ettore Randi ·
Fermin Urra · Benjamín J. Gómez-Moliner

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Abstract The use of non-invasive genetic sampling (NGS) has become increasingly popular in wildlife research but needs well-planned sampling strategies and reliable laboratory protocols. In this study, we planned to assess the reliability and success of species and individual identifications of sympatric martens (European pine marten *Martes martes* and stone marten *Martes foina*) by genotyping non-invasively collected faecal samples. First, we developed a novel and accurate multiplex panel of 15 microsatellite loci, selected by cross species amplification of 41 loci. The application of this panel facilitated species distinction, discarding the presence of putative hybrids. Then, we assessed the impact of sample collector skills on the lab protocol performances. The faecal

DNA quality was evaluated by (a) the success of polymerase chain reaction–restriction fragment length polymorphism identification of the two *Martes* species and (b) the genotyping success and error rates of individual pine marten identifications. The survey was conducted over all the sympatric range of the two species in the Iberian Peninsula by three groups of sample collectors with different experience: expert wildlife biologist, trained volunteers and technical staff from natural parks. Results show that the different expertise between sample collectors significantly influences the success rate of pine marten individual genotyping, but not the species identification success rate. Based on our results, we recommend conducting sampling by experienced field biologist to maximise the quality of NGS and ensure accurate genotyping success. Application of our methods to field collected scats can be used in a cost-effective way to investigate distribution, patterns of genetic diversity and structure as well as to estimate population abundance for sympatric martens.

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A. Ruiz-González · M. J. Madeira · B. J. Gómez-Moliner
Department of Zoology and Animal Cell Biology, University of the Basque Country, (UPV/EHU), C/ Paseo de la Universidad 7, 01006 Vitoria-Gasteiz, Spain

A. Ruiz-González (✉) · M. J. Madeira · B. J. Gómez-Moliner
Systematics, Biogeography and Population Dynamics Research Group, Lascaray Research Center, University of the Basque Country, (UPV/EHU), Avda. Miguel de Unamuno 3, 01006 Vitoria-Gasteiz, Spain
e-mail: aritz.ruiz@ehu.es

E. Randi
Laboratorio di genetica, Istituto Superiore per la Protezione e la Ricerca Ambientale (ISPRA), Via Cà Fornacetta 9, 40064 Ozzano dell'Emilia, Bologna, Italy

F. Urra
Gestión Ambiental de Navarra, S.A.,
S.A. Padre Adoain, 219–Bajo,
31015 Pamplona, Navarre, Spain

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Introduction

The European pine marten (*Martes martes*) and the stone marten (*Martes foina*) live to a large extent sympatrically over a wide area of Europe (Proulx et al. 2004). The North Iberian Peninsula currently represents both the southwestern distribution limit of the pine marten range and the southernmost sympatric area with the stone marten (López-Martín 2007). The pine marten is either threatened or scarce

in many countries, where forest habitat loss and fragmentation are major threats (Bright 1993; Mergey et al. 2011). By contrast, the number of stone martens has increased in many countries (Proulx et al. 2004), in some of which they have even invaded urban areas (Herr et al. 2009). These contrasting demographic trends and probable inter-specific competition may lead to the contraction of the pine marten range and the geographic expansion of the stone marten (Goszczynski et al. 2007). Hence, it is important to assess the presence and estimate the abundance of these species in different parts of Europe (Proulx et al. 2004). However, there is considerable difficulty in studying and monitoring their populations, given our limited knowledge of their ecological niches in sympatric areas, their extremely elusive behaviour and the fact that their faeces are morphologically indistinguishable (Ruiz-Gonzalez et al. 2008).

Molecular phylogenetic studies showed that *Martes martes* and *Martes foina* are closely related species included in the subgenus *Martes* (Stone and Cook 2002; Koepfli et al. 2008). Within subgenus *Martes*, the fossil record and molecular data indicate an earlier divergence for *Martes foina* (2.8–3.1 mya; Koepfli et al. 2008) with regard to the other extant members in this subgenus (*Martes martes*, *Martes zibellina*, *Martes americana*, *Martes melampus*) (Anderson 1970; Stone and Cook 2002; Koepfli et al. 2008). Although hybridisation between some *Martes* species has been documented (Davison et al. 2001; Kyle et al. 2003), sympatric *Martes martes* and *Martes foina* do not seem to interbreed, or at least this has not been previously reported (Davison et al. 2001). However, taking into account their close phylogenetic relationship (Stone and Cook 2002; Koepfli et al. 2008) and the documented hybridisation events between other true marten species, additional surveys and analyses, including population genetic-level sampling, will be very valuable in confidently resolving relationships among these recently evolved species (Koepfli et al. 2008) and verifying possible hybridisation processes.

Molecular methods incorporating non-invasive sampling (NGS) via the collection of scats or hairs are currently widely used for population monitoring of elusive carnivores (Waits and Paetkau 2005; Schwartz and Monfort 2008; Beja-Pereira et al. 2009). The standard method for monitoring marten populations is scat transect surveys (Birks et al. 2004). Consequently, faeces should provide a plentiful non-invasive source material that is easy to collect for genetic surveys (Ruiz-Gonzalez et al. 2008). However, the applicability of non-invasive techniques to a new species must be explored before embarking on large-scale studies (Taberlet and Luikart 1999; Valière et al. 2007). The drawbacks of non-invasively collected faecal samples of mustelids are (a) that species determination of faecal samples may be misidentified even by experienced collectors (Davison et al. 2002; Birks et al. 2004) and (b) extracts from field-collected faeces usually yield

little target DNA and may contain polymerase chain reaction (PCR) inhibitors, potentially leading to false allele (FA) amplification and/or allelic dropout (ADO) on microsatellite genotyping procedures (Waits and Paetkau 2005). Consequently, working with non-invasively obtained faecal samples of sympatric martens requires a previous determination of species identity so that we can thereafter apply a valuable and informative microsatellite panel useful on faecal samples. In this context, previous studies have recommended that any study requiring a non-invasive genetic method should be preceded by a pilot study as to assess both the probability of identity and genotyping errors (Valière et al. 2007).

Recent progress in molecular techniques has supplied several non-invasive genetic methods for the species identification of martens and related mustelids (e.g. Gomez-Moliner et al. 2004; Lucentini et al. 2007; O'Reilly et al. 2008; Ruiz-Gonzalez et al. 2008; Oliveira et al. 2010), which can be applied to field monitoring. Nonetheless, a suitable microsatellite panel for reliable faecal DNA genotyping of both marten species is unavailable, which could provide valuable information for population censuses and to address questions regarding the population genetics of free ranging marten populations.

Previous studies have given information about the genetic variability and structure of several European populations of the pine marten (Kyle et al. 2003; Pertoldi et al. 2008; Mullins et al. 2010; Colli et al. 2011; Schwartz et al. 2012; Mergey et al. 2012). However, information regarding a valuable microsatellite panel for Iberian marten populations is still unavailable. Moreover, these studies were mainly focused on tissue (Kyle et al. 2003; Pertoldi et al. 2008; Colli et al. 2011; Mergey et al. 2012) or remotely plucked hair (Mullins et al. 2010), so information about the effectiveness of microsatellite markers on faecal marten samples is currently unavailable.

In addition, to effectively and efficiently apply faecal DNA analysis in large-scale studies, it is important to identify the variables that impact PCR amplification success (Murphy et al. 2007). Although laboratory procedures can be optimised to reduce the impact of genotyping errors and increase genotyping success [e.g. faecal preservation method (Murphy et al. 2002), DNA extraction method (Goossens et al. 2000; Frantz et al. 2003; Wehausen et al. 2004) and amplification method (Goossens et al. 2000; Piggott et al. 2004)], extrinsic field factors [e.g. age of the faecal sample (Piggott 2004; Santini et al. 2007; Brinkman et al. 2010), weather conditions (Piggott 2004; Murphy et al. 2007; Brinkman et al. 2010), diet (Murphy et al. 2003) and/or season (Hajkova et al. 2006)] are usually difficult to control in non-invasive field surveys (Beja-Pereira et al. 2009; Monterroso et al. 2012). Also, the considerable remaining variability among studies implies that other unidentified parameters are acting (Broquet et al. 2007; Monterroso et al. 2012). In non-invasive genetic studies, samples are often

collected by staffs widely varying in field experience [e.g. game wardens and hunters (Bellemain et al. 2005), trained field volunteers (Jacob et al. 2010), specialised wildlife biologists (Rosellini et al. 2008), rangers and technical staff from protected areas and/or a combination of different personnel (Lucchini et al. 2002)]. The difference in the level of expertise between sample collectors potentially impacts not only the (a) accuracy in species identification of faecal samples (Davison et al. 2002; Prugh and Ritland 2005) but also (b) the effectiveness in collecting valuable fresh samples in order to maximise the DNA quality for genetic analysis (i.e. mitochondrial DNA (mtDNA) species identification and microsatellite genotyping), as success rates are greatly improved by the use of very fresh samples (Piggott 2004; Murphy et al. 2007; Santini et al. 2007). However, this bias may be limited if volunteers have the opportunity to train in species identification and fieldwork (Prugh and Ritland 2005; Jacob et al. 2010). In spite of the clear relevance of this field parameter for reliable and cost effective non-invasive genetic studies, to our knowledge, the effect of sample collector experience on species identification via mitochondrial DNA and on microsatellite genotyping success and errors rates has not previously been evaluated.

Thus, the aims of this study were (a) to develop a reliable and accurate multiplex panel of microsatellites for the individual identification of *Martes martes* and test its applicability to *Martes foina* in both tissue and scat samples, (b) to apply this panel for the detection of possible hybridisation events between both species in sympatric areas, (c) to assess the impact of sample collectors field experience on the success of polymerase chain reaction–restriction fragment length polymorphism (PCR-RFLP) identification of *Martes* sp. faecal samples and (d) on genotyping success and the error rates of pine martens faecal samples by conducting a genetic survey across the sympatric area of both marten species in the northern Iberian Peninsula.

Materials and methods

Sample collection and DNA extraction

Tissue samples

Fresh tissue samples from road-killed animals were used to isolate DNA from the target marten species, *Martes martes* ($n=35$) and *Martes foina* ($n=35$). Five specimens from each species were collected in different sampling locations (sampling areas 1–7) distributed across a wide geographic area in the northern Iberian Peninsula which covers the sympatric distribution range of the two species (López-Martín 2007)

(Fig. 1). DNA was isolated using the DNeasy® Tissue DNA extraction kit (Qiagen), according to the manufacturer's instructions.

Faecal samples

In the same area, sampling for faeces was conducted between 2006 and 2009. To assess the impact of sample collector experience on the results of microsatellite genotyping and species identification rates, samples were collected opportunistically by three different kinds of staff: (a) wildlife biologists with recognised experience in non-invasive surveys of carnivores (WB), (b) volunteers who received a training course (a field course of 2–3 days) on how to identify fresh *Martes* sp. scat samples (TV) and (c) rangers and technical staffs from different national and regional park services (TS). The technical staff was not specifically trained in collecting *Martes* samples, because this staff is supposed to have enough skills to conduct carnivore surveys. Before sampling, we mailed a detailed description of the project, the sampling protocol and material for sample collection to each of the collaborators. The wide study area and long-term programme did not permit the standardisation of sampling efforts across space and time. However, the proportion of samples obtained from each staff was similar in each season.

Martens and other carnivores use forest roads and frequently defecate on them as a way of visual-scent marking (Birks et al. 2004). Thus, sampling was conducted systematically along linear features, such as forest trails and paths. Morphological characteristics, such as size and shape, were used to presumably distinguish the faeces of the genus *Martes* from those of other medium-sized carnivores such as the red fox and the wildcat (Ruiz-Gonzalez et al. 2008).

For each faecal sample, sampling date, sample collector and Universal Transversal Mercator coordinates using a global positioning system (Garmin eTrex) were recorded. Collaborators were warned to collect only fresh samples (<5 days). Fresh faeces were characterised by the presence of a mucus layer and lack of any sign of dehydration (Rosellini et al. 2008). Moreover, to assess the validity of the fresh designation of each faecal sample, a preselection procedure was applied by evaluating a photograph of each faecal sample, discarding samples typified as fresh that presumably not correspond to this category.

A portion of each “marten-like” faecal sample was picked up by a mono-use stick and preserved in individually labelled 2-mL autoclaved tubes containing 96 % ethanol and preserved at room temperature until processed (Ruiz-Gonzalez et al. 2008).

Special care was taken to avoid the risk of cross-contamination (Taberlet et al. 1999; Pompanon et al. 2005), performing all faecal DNA extractions in a dedicated

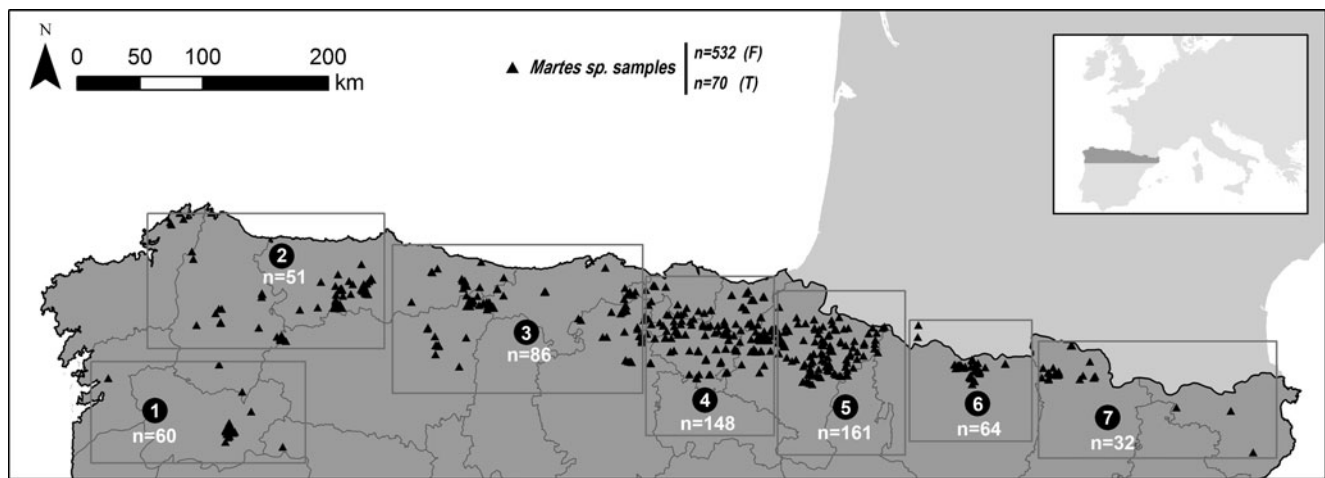


Fig. 1 GIS-determined geographical locations and number of samples ascribed to the target species after applying the PCR-RFLP method to faecal (F) ($n=532$) and tissue samples (T) ($n=70$)

collected from the entire sympatric distribution range in the northern Iberian Peninsula. Sampling areas are designated by successive numbers (1–7)

laboratory on a sterile bench, using filter tips throughout. To monitor potential contamination, we included one negative extraction control per extraction process. DNA was then extracted using the QIAamp® DNA Stool Mini Kit (Qiagen) following the manufacturer's protocol for DNA extraction from stool samples. The DNA was eluted from the silica membrane twice, using 100 μ L buffer AE (Qiagen).

Species identification

Species identification was accomplished by the PCR-RFLP method, providing for an effective genetic identification of sympatric marten species (Ruiz-Gonzalez et al. 2008). A small fragment (276 bp) of mtDNA D-loop region was amplified by PCR with specific primers developed for *Martes* and *Mustela*. The primers were designed to amplify small fragments to maximise the probability of amplification of degraded DNA. Following DNA amplification, PCR products were digested with the restriction enzymes *Hae*III and *Rsa*I. The combined use of both enzymes produced a species-specific banding pattern allowing the scats of *Martes martes* and *Martes foina* to be discriminated (see Ruiz-Gonzalez et al. 2008 for further details).

Microsatellite analysis

Development of a microsatellite multiplex protocol for Martes martes and Martes foina useful for faecal DNA genotyping

Forty one microsatellite loci identified in the genomic DNA of other mustelid species were tested for cross-species amplification and polymorphism in *Martes martes* and *Martes*

foina. The 41 microsatellite primer sets were originally developed for *Martes americana* (Ma1, Ma2, Ma3, Ma4, Ma5, Ma8, Ma9, Ma10, Ma11, Ma15, Ma18 and Ma19; Davis and Strobeck 1998), *Gulo gulo* (Gg7 and Gg14, Davis and Strobeck 1998; Gg454, Walker et al. 2001), *Meles meles* (Mel1 and Mel6, Bijlsma et al. 2000; Mel10, Domingo-Roura et al. 2003), *Mustela lutreola* (MLUT27 and MLUT04, Cabria et al. 2007), *Neovison vison* (Mvis020, Mvis022, Mvis072 and Mvis075, Fleming et al. 1999; Mvi39 and Mvi57, O'Connell et al. 1996), *Mustela erminea* (Mer 22, Mer 41 and Mer 95; Fleming et al. 1999), *Lutra lutra* (Lut604 and Lut615, Dallas and Piernney 1998) and *Martes pennanti* (MP0059, MP0084, MP0100, MP0144, MP0175, MP0188, MP0200, MP0234, MP0247 and MP0288; Jordan et al. 2007). These loci were selected for their high polymorphism and heterozygosity, as shown in previous studies of pine martens (Kyle et al. 2003; Pertoldi et al. 2008) or related species (Jordan et al. 2007). Recently developed specific microsatellite markers for *Martes martes* (Natali et al. 2010) and *Martes foina* (Basto et al. 2010) were not tested as they were not available during the execution of this research.

PCR conditions were firstly optimised for each primer in singleplex and the annealing temperature was selected after temperature gradient PCR on tissue-derived DNA. Microsatellite amplifications were performed in a total volume of 15 μ L with 2 μ L DNA, 0.2 μ L of each primers (20 pmol/ μ L), 1.7 μ L 10 $\times$ reaction buffer, 0.64 μ L MgCl₂ (50 mM), 0.5 μ L dNTP (2.5 mM), 0.1 μ L of BSA (10 mg/mL), 11.56 μ L of sterile water and 0.1 μ L *Taq* polymerase (5 U/ μ L). The following PCR conditions were used for all amplifications: after incubation for 5 min at 94 $^{\circ}$ C, the samples were subjected to 40 amplification cycles in a BioRad iCycler (Version 3.021, Bio-Rad Laboratories)

consisting of denaturation for 1 min at 94 °C, annealing for 1 min at 54–59 °C and a final extension stage of 1 min at 72 °C.

Loci with a low rate of amplification, monomorphic in both species, and with a high quantity of unspecific amplification were discarded. This first screening was evaluated with 15 tissues and 10 faecal samples of each species from the Iberian Peninsula (unreported data).

From the 41 screened loci, we selected a suite of 15 markers to design a multiplex protocol with special emphasis on demonstrated variability in pine martens, amplification strength with small quantities of DNA (faecal DNA) and powerful enough to allow individual identification. Forward primers were fluorescently labelled with different dyes (6-FAM, NED, PET and VIC) to enable multiplex electrophoresis of microsatellite products. In order to avoid noise from variable adenylation during the PCR, the “pig-tail” sequence GTTCTT was added to the 5'-end of each reverse primer (Brownstein et al. 1996). PCR multiplex amplifications were carried out with QIAGEN Multiplex PCR kits using the manufacturer's protocol in a total volume of 10 µL with 2 µL of DNA and 2 pmol of each fluorescently labelled forward and unlabelled reverse primers. We applied a hot-start thermocycling protocol. The initial polymerase activation (Hot Start PCR) was done at 95 °C for 15 min, followed by 42 cycles (35 for tissue samples) of denaturation at 94 °C for 30 s, primer annealing at 57 °C for 90 s and sequence extension at 72 °C for 60 s, and a final extension step at 60 °C for 30 min.

In addition to the negative controls for extraction, negative PCR controls were included as proposed by Pompanon et al. (2005). Additionally, we amplified a reference sample as a positive control and test if the electrophoretic mobility of the fragments was consistent across runs (Davison and Chiba 2003).

In non-invasive genetic sampling, genotyping errors occur due to increased rates of null alleles, ADO and FA (Taberlet et al. 1999; Pompanon et al. 2005). Therefore, we followed a multiple-tube approach from Taberlet et al. (1996), amplifying each DNA extract in four replicates and in separate rooms dedicated to low DNA-content samples.

First, to test the applicability of the multiplex protocol designed on non-invasive genetic surveys, 21 PCR-RFLP-identified faecal samples of each species were genotyped four times per locus to estimate error rates and genotyping success. These samples were randomly chosen from each kind of sample collector (seven from WB, seven from TV and seven from TS). Moreover, 10 out of the 35 tissue samples of each species were amplified four times per locus to estimate error rates for tissue samples. As errors were almost absent from tissue samples (see “Results”), the remaining 25 tissue samples were amplified only once per locus.

Later, in order to assess genotyping success and error rates for each sample collector, all the faecal samples identified by the PCR-RFLP method as pine marten were genotyped at 15 loci. DNA quality was initially screened by PCR-amplifying each DNA sample four times at four loci (multiplex 1: MP0188, MP0059, Gg-7 and Ma1). Multiplex 1 was selected as quality screening, since the results obtained for this for loci are indicative of the genotyping success for the full panel of 15 microsatellites (data not shown). Only samples showing >50 % positive PCRs were further amplified four times for the remaining 11 loci. Samples with ambiguous results after four amplifications per locus or with <50 % successful amplifications across loci were removed from further analyses as they were not considered reliable genotypes.

Multiplex PCR products were run on an ABI (Foster City, CA) 3130XL automated sequencer (Applied Biosystems), with the internal size standard GS500 LIZ™ (Applied Biosystems). Fragment analyses were conducted using ABI software GeneMapper 4.0.

Data analysis

Probability of identity, genotype checking and individual identification

To test the discrimination power of the set of 15 microsatellites, we computed the probability of pairs of individuals bearing an identical multilocus genotype (P_{ID}) with GIMLET v 1.3.4 (Valière 2002), using a data set of genotypes obtained from tissue DNA of 35 pine and stone martens, respectively. P_{ID} calculations were performed with both the unbiased equation for small sample size and the equation for siblings. The more conservative P_{ID} for full sibs (P_{ID-sib}) was estimated as an upper limit to the probability that pairs of individuals would share the same genotype. As advocated in Waits et al. (2001), the observed probability in the population ranged from unbiased P_{ID} to P_{ID-sib} (Waits et al. 2001).

Consensus genotypes from four replicates of faecal DNA samples were reconstructed using GIMLET v 1.3.4 (Valière 2002), accepting heterozygotes if the two alleles were seen at least in two replicates and homozygotes if a single allele was seen at least in three replicates. Afterwards, we grouped identical consensus genotypes using the same software, in order to identify the number of different individuals in our data set. GIMLET was also used to estimate PCR success and genotyping errors: ADO (when a heterozygote from the consensus genotype is typed as a homozygote from repeated genotypes) and FA (when a homozygote from the consensus genotype is typed as a heterozygote) (Pompanon et al. 2005).

Genetic diversity indices were calculated using a data set of genotypes obtained from the 70 tissue DNA (Mm=35, Mf=35) and the 42 faecal DNA samples used for the microsatellite panel optimization procedure (complete individual genotypes: Mm=12, Mf=13).

We used Genetix (Belkhir et al. 2004) to estimate observed (H_O) and expected heterozygosities (H_E), the number of total alleles (N_A), shared alleles (S_A) and private alleles (P_A) for each locus and for each of the tested species in this study. FSTAT v 2.9.3.2 (Goudet 1995, 2001) was used to test if differences in H_O values were significant between the pine marten and the stone marten.

Deviation from Hardy–Weinberg equilibrium (HW) was tested using the exact test implemented in GENEPOP version 4.0 (Raymond and Rousset 1995; Rousset 2008). Statistical significance was evaluated by running a Markov chain Monte Carlo (MCMC) consisting of 10,000 batches of 10,000 iterations each, with the first 10,000 iterations discarded before sampling (Guo and Thompson 1992). Significance levels were adjusted with sequential Bonferroni correction in order to correct for the effect of multiple tests (Rice 1989). MICRO-CHECKER software (Van Oosterhout et al. 2004) was used to check for potential scoring errors and the presence of null alleles.

Bayesian admixture analysis of sympatric martens

STRUCTURE 2.2 (Pritchard et al. 2000; Falush et al. 2003) was used to determine the level of admixture, calculated as the proportion of the genome of an individual that originated from each of the two parental species. Simulations were run using a burn-in period of 10^5 sweeps followed by 10^6 MCMC iterations. Independent runs of K (i.e. number of clusters or gene pools assumed) were performed from one to four clusters and repeated 10 times to check for consistency in the results. Admixture ancestry and correlated allele frequency models were used without prior knowledge of genetic information. Individual assignment probabilities refer to the proportion of the individual's genome which originated from European pine marten (q_i) or stone marten (q_j). First, 70 tissue samples of both marten species of known phenotype and identified by the PCR-RFLP method (Mm=35, Mf=35) were run in STRUCTURE in order to verify if the designed multiplex panel correctly discriminates the two species. Later, faecal DNA samples identified by the PCR-RFLP method and with a complete genotype (Mm=126, Mf=13) were analysed to detect possible hybridisation events in non-invasively collected samples.

Moreover, we calculated pairwise F_{ST} values (Weir and Cockerham 1984) and tested for pairwise genetic differences among clusters identified by STRUCTURE using a randomisation procedure implemented in FSTAT v 2.9.3.2 (Goudet 1995; Goudet 2001). Statistical significance at the

0.05 level was evaluated after the Bonferroni correction for multiple comparisons (Rice 1989).

Effect of sample collector experience on species identification and genotyping success

In order to evaluate the impact of the field experience of sample collectors on faecal DNA sample quality for genetic analysis, we evaluated five different parameters for each of the three kinds of sample collectors: (1) *Martes* sp. identification rate (proportion of samples identified by the PCR-RFLP method as *Martes martes* or *Martes foina*: Msp), (2) microsatellite amplification rate (PCR replicates that yielded at least one scoreable allele for full multilocus microsatellite genotypes: PCR+), (3) genotyping error rates (FA and ADO), (4) the proportion of samples that did not pass the microsatellite quality screening (SCR−) and (5) the proportion of samples yielding a reliable genotype (samples correctly genotyped at 15 loci: G+). The first parameter was calculated from all the analysed samples and the other four from all the faecal samples identified as pine marten.

To compare the proportions obtained for each parameter and to determine whether they can be considered as equal for the different sample collectors or if at least two proportions are significantly different, we used a chi-square test based on 10,000 Monte Carlo simulations. The Marascuilo procedure (Marascuilo and Slaughter 1981) was also used if the Monte Carlo simulation rejected the null hypothesis, as the Marascuilo procedure compares all pairs of proportions, which enabled the pairwise differences to be identified. Statistical analysis was performed using XLSTAT (Addinsoft™, NY, USA). Results were considered statistically significant at $P=0.05$ level.

Results

Reliability of a microsatellite multiplex panel for individual identification of *Martes martes* and *Martes foina*: genetic variability, probability of identity and genotyping errors

Details of the multiplex panel of 15 microsatellites loci, selected by cross species amplification of 41 loci, are shown in Table 1. All but three of the analysed microsatellite loci (MP0188, Mel-10 and Ma-19 were monomorphic in *Martes foina*) were polymorphic in both mustelid species, with an average number of alleles per polymorphic locus of 4.13 and 3.31 (ranging from 2 to 6) in *Martes martes* and *Martes foina*, respectively (Table 1). Only 13 out of the 96 alleles found in both species (13.54 %) were shared alleles, whereas 33 (34.38 %) and 50 (52.08 %) were private alleles for *Martes martes* and *Martes foina*, respectively. Interestingly,

Table 1 Properties of the 15 microsatellite multiplexed loci used in this study and summary of the genetic variability assessed per locus and species. The table includes number of alleles (N_A), shared alleles (S_A), private alleles (P_A), observed (H_O) and expected (H_E) heterozygosities, rates of positive PCR (PCR+), dropout (ADO) and false allele (FA) for each locus and for each of the species tested in this study (Mm=*Martes martes*, Mf=*Martes foina*)

Multiplex	Locus	Dye	Size range		N_A		S_A		P_A		H_E		H_O		PCR+		ADO		FA	
			Mm	Mf	Mf	Mm	Mf	Mm	Mf	Mm	Mf	Mm	Mf	Mm	Mf	Mm	Mf	Mm	Mf	Mm
MULT_1	Gg-7	PET	164–172	166–174	3	5	2	1	3	3	0.342	0.713	0.371	0.638	0.13	0.96	0.333	0.143	0	0
	Ma-1 *	NED	204–210	203–213	6	4	0	6	4	4	0.675	0.520	0.629	0.444	0.94	0.91	0.345	0.167	0	0
	MP0059*	VIC	140–148	139–147	4	4	0	4	4	4	0.707	0.615	0.628	0.472	0.96	0.98	0.150	0.092	0	0.072
MULT_2	MP0188	6-FAM	113–121	105	1	3	0	1	3	3	–	0.483	–	0.418	0.81	0.97	0	0.139	0	0
	Lut-433	6-FAM	108–112	102–108	3	3	1	2	2	2	0.292	0.513	0.228	0.5000	1	0.96	0.400	0.093	0	0.033
	Mel-1	PET	265–275	264–272	3	5	0	3	5	5	0.378	0.643	0.371	0.667	0.98	0.87	0.281	0.174	0	0.009
MULT_3	Mel-10	NED	158–166	127	1	5	0	1	5	5	–	0.536	–	0.472	1	0.88	0	0.375	0	0.014
	Lut-435*	VIC	129–143	135–139	3	3	1	2	2	2	0.493	0.508	0.543	0.361	0.94	0.86	0.333	0.114	0.167	0.027
	Ma-19	PET	206–210	211	1	3	0	1	3	3	–	0.578	–	0.556	0.96	0.85	0	0.192	0	0
MULT_4	Mvi-57	6-FAM	100–114	100–108	4	5	2	2	3	3	0.484	0.416	0.543	0.444	1	0.94	0.300	0.167	0	0.014
	Mvi072	NED	260–276	264–270	4	6	2	2	4	4	0.477	0.726	0.457	0.806	0.94	0.77	0.190	0.183	0	0.023
	Lut-615	6-FAM	111–115	113–119	4	2	1	3	1	3	0.469	0.490	0.400	0.528	0.88	0.89	0.389	0.269	0	0.014
MULT_5	Ma-2	NED	167–179	169–177	4	5	2	2	3	3	0.629	0.672	0.628	0.583	0.85	0.93	0.167	0.274	0	0.008
	Mer41	VIC	150–160	156–164	3	5	2	1	4	4	0.277	0.757	0.286	0.805	0.94	0.87	0.361	0.316	0	0
	Mlut-27	PET	198–204	186–194	2	4	0	2	4	4	0.408	0.616	0.457	0.583	0.94	0.91	0.325	0.333	0	0
Total					46	62	13	33	50						0.88	0.90	0.238	0.202	0.011	0.014
Mean					3.31	4.133					0.376	0.586	0.369	0.552						

Means for the number of alleles and heterozygosities are computed over polymorph loci

* $P=0.05$ (loci deviated from Hardy–Weinberg proportions at this level (Ma-1 for *Martes foina*; MP0059 and Lut-435 for *Martes martes*))

alleles found at seven microsatellite loci were species specific with slight differences in allele size (Table 1).

Observed heterozygosity was lower than expected heterozygosity in both species. The mean expected heterozygosity over all polymorphic loci was 0.586 and 0.376 and the mean observed heterozygosity was 0.552 and 0.369 for *Martes martes* and *Martes foina*, respectively (Table 1). Comparison of H_O revealed significant differences between pine martens and stone martens ($P < 0.05$).

Significant departure from HW equilibrium after Bonferroni correction was observed within each of the study species ($P < 0.05$). Two loci showed significant departure from HW equilibrium in pine martens (MP0059 and Lut-435), whereas only one locus deviated from HW equilibrium proportions in stone martens (Ma-1). The results of MICRO-CHECKER rejected the hypothesis that putative null alleles were causing bias in genetic diversity and population differentiation estimates.

Based on 15 microsatellite data, high genetic differentiation was detected between the two mustelid species ($F_{ST} = 0.485$, $P < 0.001$).

Probability of identity was calculated on the basis of allele frequencies estimated from the 35 tissue samples of each species. The overall P_{ID} using all loci for pine martens and stone martens was 5.42×10^{-11} and 5.42×10^{-7} and the overall P_{ID-sib} was 4.47×10^{-5} and 1.95×10^{-3} , respectively. When locus Gg-7 was excluded, being difficult to type locus for stone marten faeces samples, the overall P_{ID} became 1.21×10^{-6} and the overall P_{ID-sib} was 2.79×10^{-3} . Thus, P_{ID-sib} was again lower than the 0.01 threshold necessary to prevent the shadow effect, i.e. the presence of two or more individuals with the same multilocus genotype (Mills et al. 2000) (Fig. 2). Thus, the 15 loci used in this study were necessary yet sufficient, to distinguish with 99 % certainty between sibling pine or stone martens (Mills et al. 2000) (Fig. 2). Moreover, the P_{ID-sib} showed that the proportion of individuals with identical profiles greatly decreases if the most informative loci were used (three loci in *Martes martes* and eight loci in *Martes foina*) (Fig. 2).

Genotyping errors from tissue were estimated from 10 samples of each marten species. No ambiguous genotypes were detected. The proportion of PCR + s varied between 90 and 100 % among loci and from 86 to 100 % among samples for both species. Only one and three ADOs (< 0.001 %) were detected in pine and stone martens, respectively, at locus Gg-7 for both species.

Genotyping success and error rates for faeces were preliminarily estimated from 42 samples (21 of each species). Thirteen faecal samples of *Martes martes* (61.9 %) and 12 (57.14 %) of *Martes foina* were completely and successfully genotyped. The other samples showed less than 50 % amplification success or high genotyping error rates after four replications at 15 loci, so they were discarded as we were

not able to resolve consensus genotypes from repeated attempts. Gg-7 was not valuable for faecal DNA genotyping of stone martens because of the low amplification success rate (0.13).

The average proportion of positive PCRs for full multilocus microsatellite genotypes of faecal samples was 88 % for the stone marten and 90 % for the pine marten (Table 1) and varied among loci from 77 and 100 % in both species. The mean ADO rate for *Martes foina* and *Martes martes* was 0.238 and 0.202, respectively. The ADO rates were not homogenous in each of the multiplexed reactions and each of the species (see Table 1). False allele rates averaged 0.011 for stone martens and 0.014 for pine martens (Table 2).

Bayesian admixture analysis for species assignment

Analyses using STRUCTURE with both the admixture ancestry and correlated allele frequency models supported the existence of two distinct genetic clusters ($k=2$) based on the log probability of the data ($\ln [\Pr(X/K)]$) given in the model. Moreover, the modal value of ΔK (Evanno et al. 2005) was also shown at $K=2$ (data not shown). Cluster I grouped all the tissue samples identified as *Martes martes* by both phenotype and PCR-RFLP method, with an estimated average proportion of membership (Q_I) higher than 0.99. Cluster II included all the individuals identified as stone martens by phenotype and PCR-RFLP with $Q_{II} > 0.99$. Individual assignment probabilities (q_i) for faecal DNA samples identified by the PCR-RFLP methods as *Martes martes* and *Martes foina* were higher than 0.99 and were correctly assigned to the corresponding species.

Effect of sample collector field experience on *Martes* sp. identification and pine marten genotyping success of faecal DNA

Sample collection and *Martes* species identification

Out of 849 faecal samples collected from the entire sympatric distribution range of both marten species in the northern Iberian Peninsula (López-Martín 2007), 355 were collected by WB, 283 by TV and 211 by TS (Table 2); 215 (25.32 %) were finally discarded as they were not assigned as fresh samples according to the preselection procedure. The percentage of discarded samples was greater on samples collected by TS (29.38 %) than by TV (28.97 %) and WB (20.00 %), respectively ($\chi^2 = 9.155$, $P = 0.009$) (Table 2 and Fig. 3). The Marascuilo procedure indicated that the difference between the proportions of discarded samples was statistically significant between WB and TS and between WB and TV (Table 3 and Fig. 3). After the sample preselection procedure, we obtained a total number of 634 fresh faecal DNA samples suitable for genetic analysis (Table 2);

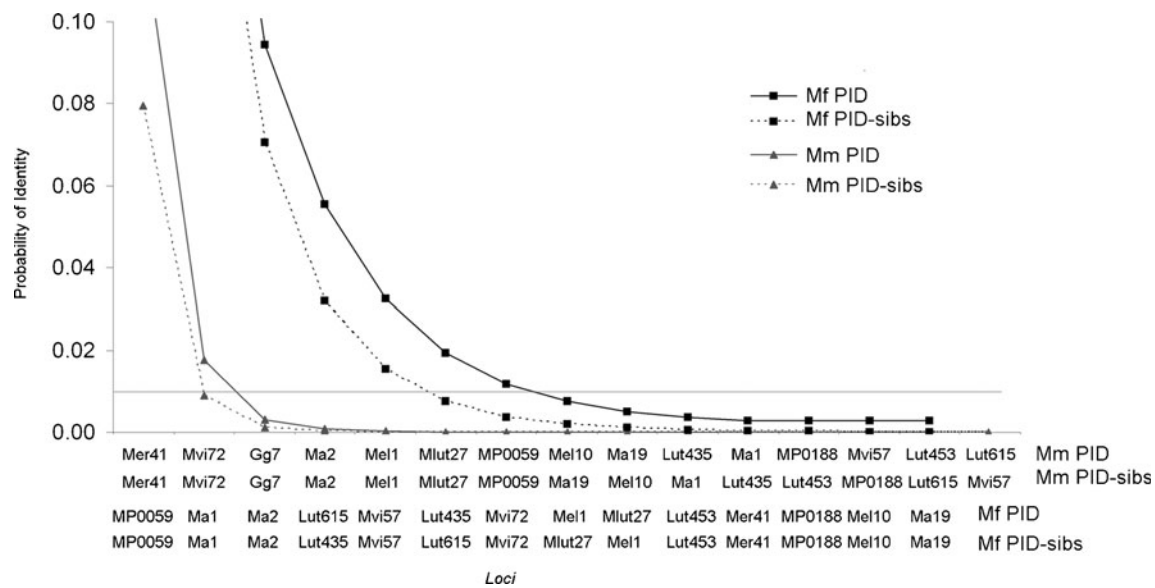


Fig. 2 Probability of identity (P_{ID}) and probability of identity for full-sibling ($P_{ID-sibs}$) values with the addition of loci in decreasing order of heterozygosity for *Martes martes* (Mm) and *Martes*

foina (Mf). The 1 % cut-off line represents the point where enough loci are typed to distinguish between individuals with 99 % certainty

532 were identified as one of the target species by our PCR-RFLP method. Thus, unequivocal species identification was possible in 83.91 % of the samples (Table 2). In the remaining 12 %, the DNA extracted was not amplified by the primers used. The geographical locations for the 532 correctly identified faecal samples in

the Iberian Peninsula are shown in Fig. 1. The species identification success rate was similar for all three sample collectors with slightly higher values for WB (85.56 %) and TV (84.58 %) in comparison with TS (79.87 %) (Table 2 and Fig. 3). However, differences were not significant ($\chi^2=2.477$, $P=0.284$).

Table 2 Number of samples collected (SC), discarded (SD) and analysed (SA); number (in brackets) and percentage of samples not amplified by mtDNA (AMP–), identified as *Martes martes* or *Martes foina* (Msp), identified as *Martes foina* (Mf), identified as *Martes martes* (Mm) and identified as *Martes martes* that failed to pass

screening (SCR–), not correctly genotyped (G–) and correctly genotyped (G+) and rates of positive PCR (PCR+), dropout (ADO) and false allele (FA) for each of the three different kinds of sample collectors: wildlife biologist (WB), trained volunteers (TV) and technical staff (TS)

		Sample collector			Total
		WB	TV	TS	
SC		355	283	211	849
	SD*	71	82	62	215
	SA	284	201	149	634
	AMP–	14.44 (41)	15.42 (31)	20.13 (30)	16.09 (102)
	Msp	85.56 (243)	84.58 (170)	79.87 (119)	83.91 (532)
	Mf	24.65 (70)	52.24 (105)	26.85 (40)	33.91 (215)
	Mm	60.92 (173)	32.34 (65)	53.02 (79)	50.00 (317)
	SCR–*	30.06 (52)	49.23 (32)	46.83 (37)	38.17 (121)
	G–	17.34 (30)	10.76 (7)	18.99 (15)	16.4 (52)
	G+*	52.60 (91)	40.00 (26)	34.18 (27)	45.43 (144)
	PCR+	0.96	0.94	0.95	0.95
	ADO	0.142	0.137	0.159	0.189
	FA	0.014	0.02	0.015	0.021
	IDIND	69	18	27	114

* $P<0.05$ (for chi-square test statistic)

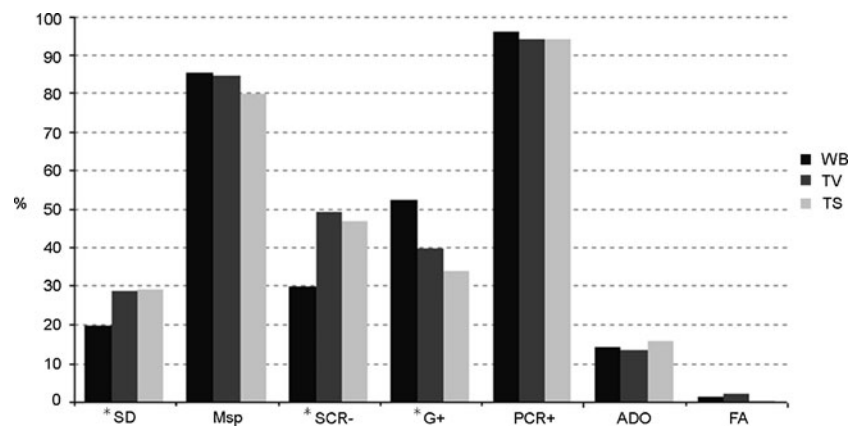


Fig. 3 Comparison of percentage of discarded samples by preselection procedure (SD), *Martes* species identification success (Msp), samples that failed to pass quality screening (SCR-), genotyping success (G+), positive PCR (PCR+), dropout (ADO) and false allele (FA) for each of the three different kinds of sample collectors: wildlife biologist (WB), trained volunteers (TV) and technical staff (TS). * p value <0.05 for chi-

square test statistic. For positive chi-square test, the Marascuilo procedure was used to determine which pairs of proportions, corresponding to each of the sample collectors, were statistically significant. For the three parameters in which the chi-square test was positive, the statistically significant pairs were SD (WB-TV, WB-TS), SCR- (WB-TS, WB-TS) and G+ (WB-TS)

Genotyping success and error rates of pine marten faecal samples

From the total scat samples identified as pine marten ($n=317$), we evaluated the effect of sample collector experience on microsatellite amplification rate (PCR+), genotyping error rates (FA and ADO), the proportion of samples that failed to pass quality screening (SCR-) and samples yielding a reliable genotype (G+). The first quality-screening test was not passed by 121 samples (38.17 %), which were immediately discarded. The proportion of the samples that did not pass quality screening was higher and statistically significant ($\chi^2=10.707$, $P=0.003$), in both TV (49.23 %) and TS (46.83 %) in comparison to WB (30.06 %) (Table 2 and Fig. 3). The Marascuilo procedure identified that the significant pairwise differences were those obtained between WB and TV and WB and TS collected samples (Table 3 and Fig. 3). The remaining 196 samples (61.83 %) were amplified for the other 11 loci. After multiple-tube genotyping, 52 samples (16.4 % of the total pine marten samples analysed) were then discarded because they showed <50 % PCR success or because of high failure rates. Full multilocus microsatellite genotypes were obtained for the remaining 144 samples (73.47 % from the samples that passed the screening and 45.43 % from the total *Martes martes* samples analysed). After a regrouping procedure, we identified 114 individual genotypes (Table 2). The highest genotyping success rate (52.60 %) was obtained for samples collected by WB, followed by TV (40.00 %) and TS (34.18 %) (Table 2 and Fig. 3). Chi-square test showed a significant difference between the proportion of samples correctly genotyped across sample collectors ($\chi^2=8.397$, $P=0.018$), and the Marascuilo procedure showed that the only significant difference was obtained between WB and TS collected samples (Table 3 and Fig. 3). The overall PCR success rate (calculated only

for correctly genotyped samples) was 95 % with no significant differences ($\chi^2=4.265$, $P=0.118$) and similar values across sample collectors (Table 2 and Fig. 3). The observed average error rates across loci were ADO=0.189 and FA=0.021. The results obtained for the three sample collectors were very similar and not statistically significant (ADO: $\chi^2=5.288$, $P=0.071$; FA: $\chi^2=1.303$, $P=0.527$) (Fig. 3 and Table 2).

From the total analysed samples ($n=634$), 16.08 % could not be assigned to species level, 33 % were identified as *Martes foina*, 8 % failed to pass quality screening and 19.08 % were not finally reliably genotyped. Thus, approximately 23 % of all the analysed samples were correctly genotyped (45.43 % if we only consider samples identified as *Martes martes* by PCR-RFLP). The global and partial results for each sample collector are summarised in Fig. 4.

Table 3 The statistically significant pairs (SD (WB-TV, WB-TS), SCR- (WB-TS, WB-TS) and G+ (WB-TS)) for the three parameters in which the chi-square test was positive

	Contrast	Value	Critical value	Significant
SD	$ p(WB) - p(TV) $	0.090	0.084	Yes
	$ p(WB) - p(TS) $	0.094	0.093	Yes
	$ p(TV) - p(TS) $	0.004	0.101	No
SCR-	$ p(WB) - p(TV) $	0.192	0.174	Yes
	$ p(WB) - p(TS) $	0.168	0.162	Yes
	$ p(TV) - p(TS) $	0.024	0.205	No
G+	$ p(WB) - p(TV) $	0.126	0.175	No
	$ p(WB) - p(TS) $	0.184	0.160	Yes
	$ p(TV) - p(TS) $	0.058	0.198	No

For positive chi-square test, the Marascuilo procedure was used to determine which pairs of proportions, corresponding to each of the sample collectors, were statistically significant

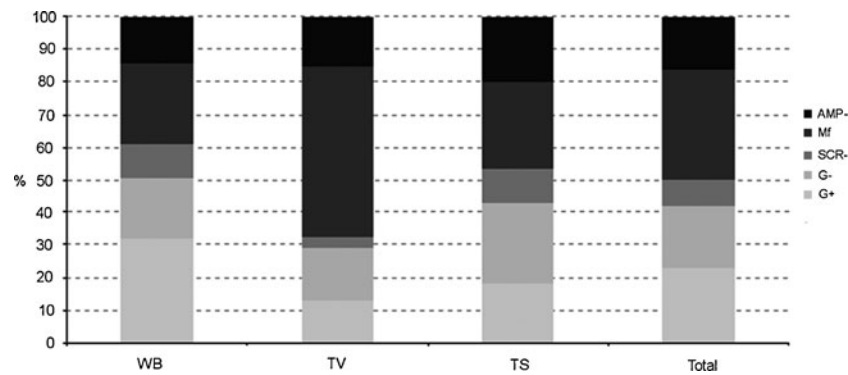


Fig. 4 Percentage of samples from the total analysed samples that were not amplified by mtDNA (AMP–), identified as *Martes foina* (Mf), identified as *Martes martes* that failed to pass quality-screening (SCR–), identified as *Martes martes* not reliably

genotyped (G–) and identified as *Martes martes* correctly genotyped (G+) for each of the three different kinds of sample collectors (WB wildlife biologist, TV trained volunteers and TS technical staff) and for the whole data set

The partial results obtained for each sample collector are slightly different and mainly biased by the differences in the percentage of samples identified as *Martes foina*.

Discussion

Research studies on non-invasive genetics of mustelids are limited and biased towards species identification methods (e.g. Gomez-Moliner et al. 2004; Lucentini et al. 2007; Ruiz-Gonzalez et al. 2008). Microsatellite genotyping of non-invasive samples has also been carried out on a limited number of mustelid species and used mainly for individual identification and population size estimation (e.g. European otter, Hajkova et al. 2009; wolverine, Hedmark and Ellegren 2007; badger, Frantz et al. 2006). To our knowledge, this study is the first to investigate and evaluate a microsatellite multiplex panel for individual identification of sympatric *Martes martes* and *Martes foina*, while also being valuable for faecal DNA genotyping. Moreover, the application of this microsatellite panel facilitated species distinction, discarding the presence of putative hybrids in sympatric populations of marten species in northern Iberian Peninsula. This paper provides also the first assay to evaluate the effect of sample collectors experience on *Martes* species identification and microsatellite genotyping success and error rates in non-invasively collected faecal samples.

Reliability of the microsatellite panel for individual identification and genetic variability assessment of sympatric martens

Reliable amplification of microsatellite markers is a prerequisite for estimating the size and structure of populations within the landscape (Valière et al. 2007). In this paper, we standardised individual genetic identification of sympatric martens from faecal DNA with protocols offering some

advantages over past efforts mainly focused on remotely plucked hair (Mullins et al. 2010). Firstly, species identification of the field-collected samples ensured that only target species samples were used for further analysis. This initial step towards species identification is crucial in landscapes where both species overlap extensively (Ruiz-Gonzalez et al. 2008). In addition, the screening of a sufficiently large number of 41 microsatellite loci for the initial assessment of variability improved our ability to ascertain the combination of loci, resulting in higher precision in probability of identity. This enabled us to get an optimal combination of 15 variable loci. Additionally, recently developed markers for *Martes martes* (Natali et al. 2010) and *Martes foina* (Basto et al. 2010) could be combined with those identified in this paper, in order to get a more comprehensive knowledge about genetic variability of both marten species in other European populations.

The most conservative estimate of P_{ID} , i.e. P_{ID-sib} statistics, suggested that DNA profiles consisting of the 15 loci used in our study would be sufficient to distinguish between individuals of each of the study species, including siblings, with >99 % certainty (Fig. 2). Therefore, they can be used in studies requiring individual identification for population size estimation based on non-invasive scat sampling.

Our success rate of pine marten microsatellite genotyping (45.43 %) is around average in comparison with many other studies undertaken on other carnivore species, which reported successful genotyping of faecal samples in the range of 14–62 % [e.g. badger *Meles meles* 62 %, Frantz et al. 2003; wolverine *G. gulo* 54 %, Hedmark and Ellegren 2007; otter *L. lutra* 14 %, Lanszki et al. 2008; 21 %, Ferrando et al. 2008; 44 %, Arrendal et al. 2007; 55–63 %, Hajkova et al. 2009; wolf *Canis lupus* 59 %, Lucchini et al. 2002; 50–61 %, Stenglein et al. 2010]. The genotyping success and error rates obtained from the 21 faecal samples from each species for the microsatellite panel optimization procedure were similar to what was obtained for the whole pine

marten survey ($n=317$). Thus, the partial results of the pilot study reflected the magnitude of the success and error rates that could be obtained on a more extensive research.

Another important issue in non-invasive genetic studies is the detection and elimination of genotyping errors, as they can dramatically affect survey results, especially when estimating abundance via genetic sampling (Valière et al. 2007). The rate of genotyping errors, therefore, should always be carefully computed and clearly reported (e.g. Broquet and Petit 2004). The frequency of allelic dropout and false alleles in the microsatellite panel optimization procedure (Mf: ADO=23.8 %, FA=1.1 %; Mm: ADO=20.2 %, FA=1.4 %) and in the whole pine marten survey (ADO=17.8 %, FA=2.9 %) was similar to that obtained in other non-invasive genetic studies of carnivores (e.g. ADO=27 %, FA=8 %, Frantz et al. 2003; ADO=14 %, FA=1 %, Hedmark and Ellegren 2007; ADO=15.8 %, FA=2 %, Ferrando et al. 2008; ADO=17.8 %, FA=2.9 %, Hajkova et al. 2009; ADO=14–43 %, FA=4 %, Stenglein et al. 2010). However, some microsatellite loci have high values of ADO rate in *Martes foina* that could be problematic on more extensive NGS studies. In this regard, an evaluation of new developed markers for this species (Basto et al. 2010) on faecal samples could provide new valuable microsatellites for faecal DNA genotyping purposes in addition of those presented in this paper.

There are a few studies focused on the genetic variability of pine martens (Kyle et al. 2003; Pertoldi et al. 2008; Mullins et al. 2010; Colli et al. 2011; Mergey et al. 2012), while no previous study is available for *Martes foina* (Schwartz et al. 2012.). Studying pine marten populations across Europe, Kyle et al. (2003) found microsatellite H_E values in the range of 0.56–0.64 for continental and 0.34–0.66 for insular populations (eight microsatellite loci). Mergey et al. (2012) and Colli et al. (2011) obtained similar values for French (eight loci, $H_E=0.48$ –0.62) and Sardinian populations (seven loci, $H_E=0.523$ –0.655), respectively. Pertoldi et al. (2008) found H_E values in the range of 0.67–0.79 in the Danish populations (11 loci), while Mullins et al. (2010) found slightly lower values of H_E for the reduced and isolated Irish populations ($H_E=0.35$, 17 loci). In this study using 15 variable loci, we detected H_E values of 0.586. So, taking into account the high number of variable loci used in this study, we can estimate that in comparison to continental European populations, the Iberian population of pine martens may have lower levels of genetic variability. However, caution is called for when trying to interpret differences in the level of genetic variability between different studies as these differences can also be attributed to the loci under study.

The lower heterozygosity levels found in *Martes foina*, in comparison to *Martes martes* (H_O , Mf=0.369; Mm=0.552), could be explained by the lower polymorphism of microsatellite markers in stone martens (Mm: $A=4.13$; Mf: $A=3.31$).

In this regard, we have to take into account that the microsatellite multiplex panel was primarily developed to be variable for pine martens as landscape genetic research on this species is ongoing (Ruiz-Gonzalez et al., in preparation). Thus, the differences observed in the main genetic variability parameters are the consequence of the set of markers used (i.e. cross-species amplification) and not due to other factors such as distribution range or effective population size.

Reliable species identification: no evidence of genetic hybridisation

The genetic analyses performed in this study were based on different molecular markers (mtDNA and nuclear DNA (nDNA)) and gave consistent results that provided detailed information regarding the reliable species identification of sympatric marten species. Microsatellite data confirmed the validity of the species identification results obtained by the PCR-RFLP method on mtDNA. The presence of species-specific microsatellite alleles and loci, together with Bayesian admixture analysis, facilitated the distinction of pure-bred species, discarding the presence of putative hybrids.

These preliminary results, together with the absence of phenotypically intermediate individuals described in the literature, indicated that there is no hybridisation between these two sympatric marten species in the Iberian Peninsula.

Effect of sample collector field experience on species identification (mtDNA) and genotyping success (nDNA) of faecal samples

Faecal samples could potentially result in highly variable DNA quality and quantity, resulting in different genotyping success and error rates (Beja-Pereira et al. 2009). As already suggested by previous non-invasive genetic studies conducted on carnivore species (Piggott 2004; Murphy et al. 2007; Santini et al. 2007), success rates are greatly improved by the use of very fresh samples. Thus, the effectiveness of sample collectors in meeting this objective is a crucial parameter.

In this study, the difference in the level of expertise among sample collectors potentially impacts the results of microsatellite genotyping, while there are no significant differences in species identification rate. Thus, while sample quality is a limiting factor for microsatellite genotyping (nDNA), it is not so for species identification rate by PCR-RFLP of mtDNA, after conducting a freshness sample pre-selection procedure.

The proportion of samples discarded from the total samples collected showed significant differences among samples collected by different personnel, demonstrating that the sampling conducted by wildlife biologists had the highest proportion of valuable fresh samples for genetic analysis.

For example, a sampling conducted by WB increased the overall available fresh sample for consecutive genetic analyses by 9 and 10 % in comparison to TV and TS, respectively.

Our results suggest that sample collector experience had no significant effect on species identification rate, suggesting that the work of non-specialised personnel may still be valuable in genetic studies focused on distribution assessment of sympatric martens as mtDNA methods can effectively work with highly variable DNA quality and quantity. This can be explained by the higher copy numbers of mtDNA compared to nDNA, conferring a greater success rate even with non-fresh samples that are not suitable for genotyping analysis. An overall high species identification rate was obtained (mean 84 %), confirming previous results of the application of the PCR-RFLP method on scat samples for monitoring the presence and distribution of sympatric marten species in the northern Iberian Peninsula (Ruiz-Gonzalez et al. 2008; Rosellini et al. 2008) and Italy (Balestrieri et al. 2010). These results are above the mean (73 %) reported in the review of non-invasive papers by Broquet et al. (2007). However, as scat identification accuracy rates are circumstance specific (i.e. potentially influenced by biological, environmental and human-related factors) (Monterroso et al. 2012), we recommend that this parameter should be always cautiously evaluated for identifying any potential source of error in carnivore surveys.

On the other hand, we demonstrated that sample collector experience can have a significant effect on nuclear DNA quality and on the ability to obtain full genotypes from pine marten scats. Surprisingly, this effect has been largely ignored in the faecal DNA literature.

Overall, if we consider the two most important indices of genotyping success (i.e. percent of samples that failed to pass the screening, SCR–, and correctly genotyped, G+), we can confirm that the level of expertise among sample collectors potentially impacts the results of microsatellite genotyping, as the proportions obtained for each parameter are not equal for the different sample collectors according to chi-square test. In this regard, WB always get the highest G+ (WB=52.60 %>TV=40 %>TS=34.18) and the lowest SCR– values (WB=30.06 %<TV 49.23 and TS=46.83) showing a clear tendency of improvement for the evaluated parameters.

The proportion of the samples that did not pass the quality screening was lower for the samples collected by wildlife biologist, who increased the overall available samples for further genotyping by 19.17 and 16.77 % in comparison to TV and TS, respectively.

Genotyping success rates (G+) also showed significant differences among samples collected by different personnel demonstrating that the degree of experience in collecting fresh quality samples increases the probability of obtaining a

correct full genotype. For example, a sampling conducted by WB increased the overall per sample finalised genotyping success by 12.6 and 18.3 % in comparison to TV and TS, respectively. However, we have to notice that the only significant pairwise difference found for the G+ is between the two most divergent values (WB, G+=52.60 % vs TS, G+=34.18), which supposes an increase of 18.3 % on G+ value. Even though the sampling conducted by WB increased the G+ value in comparison to TV (WB, G+=52.60 % vs TV, G+=40.00 %), the differences are not significant. By contrast, PCR amplification success and error rates were not significantly different among sample collectors. However, these two measures are less important indicators of differences in DNA quality as PCR amplification products alone do not necessarily reflect true genotypes (Broquet and Petit 2004). Thus, in large-scale non-invasive genetic sampling projects, the field experience of sample collectors can significantly increase the number of completed genotypes and maximise output per unit effort.

Non-invasive genetic studies may benefit from the help of different sample collectors with differences in the level of expertise, as exemplified in Bellemain et al. (2005), Jacob et al. (2010) or in our present study. However, in order to maximise non-invasive sampling and DNA quality which will insure accurate genotyping success, we recommend, whenever possible, that sampling be conducted by experienced biologists. However, due to the fact that it is not always possible to conduct large NGS studies only by experienced WB, we outline some recommendations in order to maximise the quality and the success of NGS studies developed by staff with different levels of experience: (a) First, exhaustive training activities are desirable for less experienced collectors not only before but also during the execution of the project (e.g. field training courses and workshops about NGS protocols); (b) it is advisable, even necessary, to have an active and regular communication with sample collectors during the execution of the project in order to increase their motivation and involvement in the study; (c) in this regard, economic incentives (e.g. reimbursement of expenses derived from the field work) are also desirable; and (d) additionally, we recommend evaluating the quality of the samples obtained by different staff through a preselection procedure in order to maximise output per unit effort in subsequent genetic analysis.

Applicability of species and individual identification methods for reliable population monitoring of sympatric marten species

In conclusion, the combined application of the species and individual identification methods described in this study to field collected scats can be used in a cost-effective way to

investigate species biology (including spatial distribution, patterns of genetic diversity, relatedness and population connectivity) as well as to estimate population abundance for sympatric martens in the wild, in order to identify and resolve information gaps and design effective research and management programmes, which are lacking in many European countries (Proulx et al. 2004).

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